Customer Shopping Behavior Analysis

Insights from 3,900 purchases across product categories

Dataset Overview

3,900 Transactions

Purchase records analyzed

18 Features

Data columns tracked

37 Missing Values

In review ratings

Customer Data

- Demographics (Age, Gender, Location)
- Subscription Status
- Previous Purchases

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Shipping Type

Behavior Metrics

- Discount Applied
- Review Rating
- Purchase Frequency

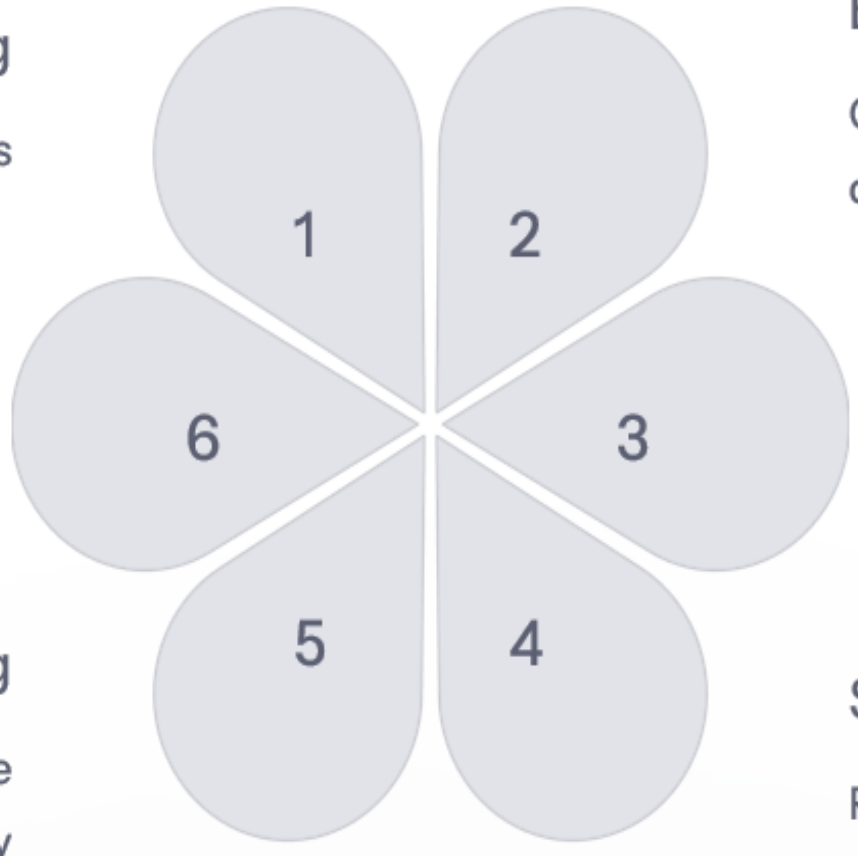
Python Data Preparation

Data Loading

Imported dataset using pandas

Exploration

Checked structure with info() and describe()



Database Load

Connected to PostgreSQL

Missing Data

Imputed missing ratings by category median

Feature Engineering

Created age groups and purchase frequency

Standardization

Renamed columns to snake case

SQL Analysis: Revenue Insights

Gender Revenue

Male: $157,890

Female: $75,191

Shipping Comparison

Standard: $58.46 avg

Express: $60.48 avg

Subscription Impact

Subscribers: 1,053 customers

Non-subscribers: 2,847 customers

Product Performance

Top Rated

Gloves (3.86), Sandals (3.84), Boots (3.82)

Discount-Heavy

Hat (50%), Sneakers (49.66%), Coat (49.07%)

Best Sellers

Blouse (171), Pants (171), Jewelry (171)

Customer Segmentation

80%

Loyal

3,116 customers

18%

Returning

701 customers

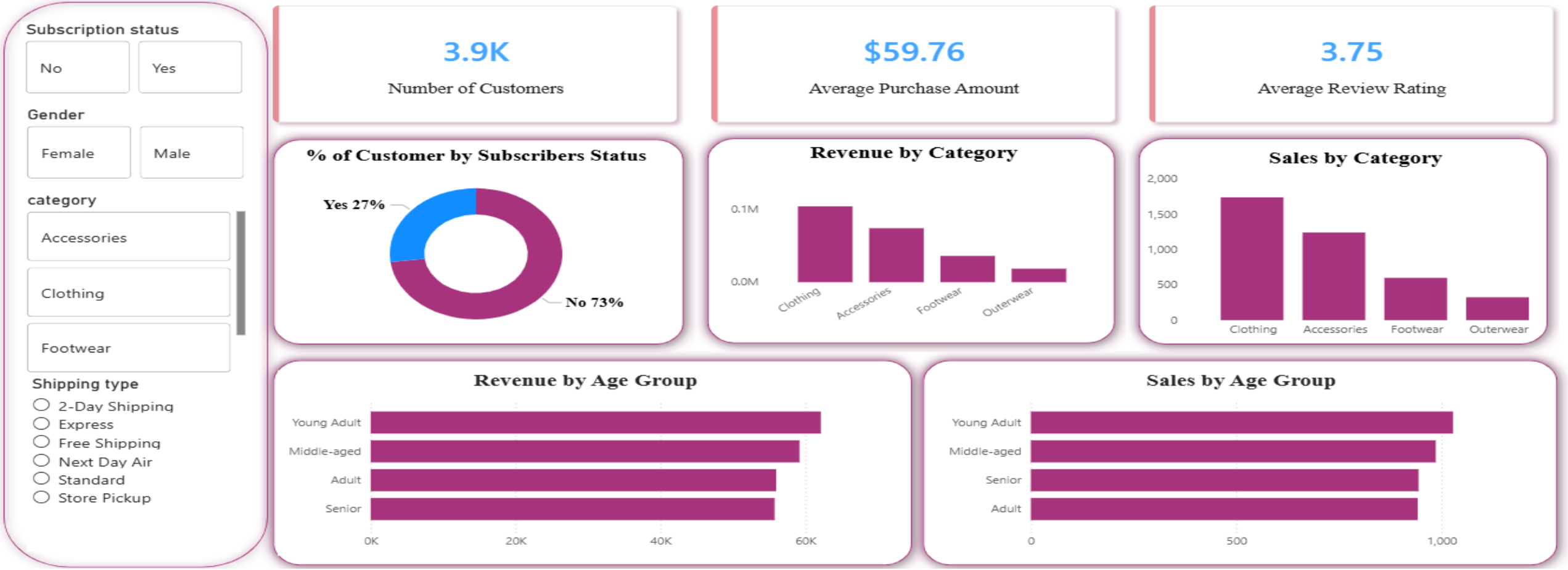
2%

New

83 customers

839 high-spending discount users identified

Customer Behavior Dashboard



Power BI Dashboard

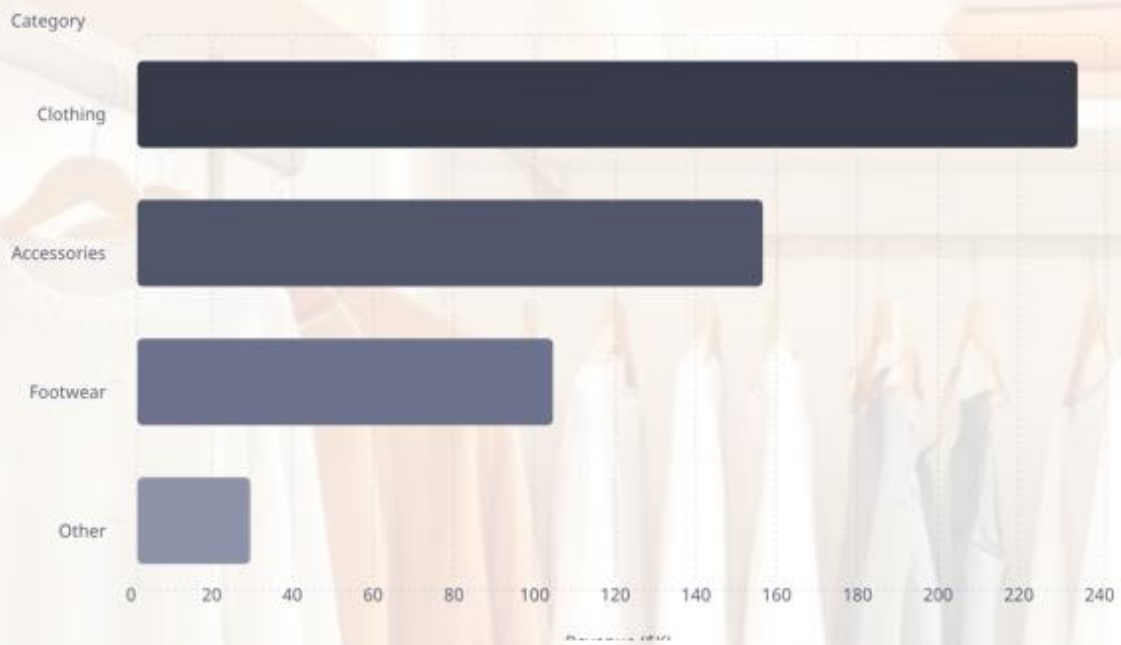
Key Metrics

- 3.9K customers
- $59.76 avg purchase
- 3.75 avg rating

Visualizations

- Revenue by category
- Sales by age group
- Subscription status

Revenue by Category



Category breakdown

- Clothing — $233K (45%)
- Accessories — $155K (30%)
- Footwear — $103K (20%)
- Other — $28K (5%)

Key insight

Clothing dominates revenue, driven by high-volume items like blouses, pants, and jackets. Strategic focus on this category can maximize ROI.

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to increase 27% subscription rate

Customer Loyalty Programs

Reward repeat buyers to move into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users